



## Práctica cero: Mecánica pulmonar

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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### Información general



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### Datos de la simulación

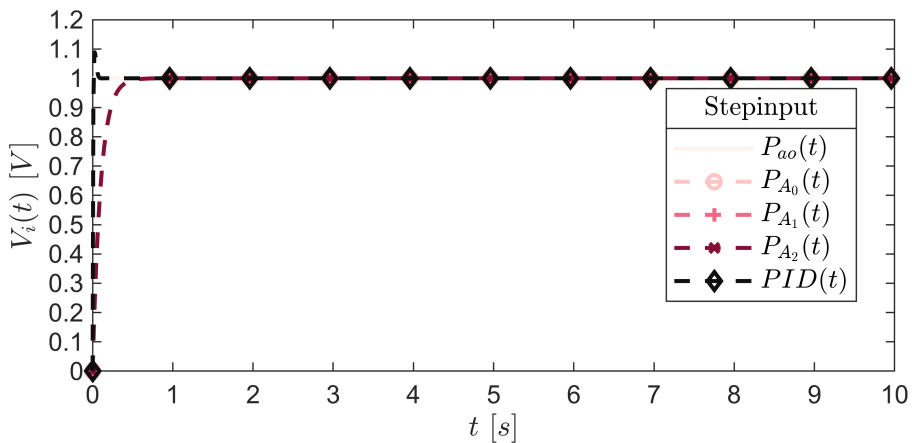
```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sistema';
open_system(file);
parameters.StopTime = tend;
parameters.Solver = 'ode15s';
parameters.MaxStep = '1E-3';
set_param('Sistema/PA0(t)', 'VectorFormat', '1-D array');
```

## Respuesta al escalón

```
Signal = 'Step';  
set_param('Sistema/S1','sw','1');  
set_param('Sistema/Pao(t)','sw','1');  
x1 = sim(file,parameters);
```

Found algebraic loop containing:  
Sistema/PA(t)/Gain  
Sistema/PA(t)/Gain1  
Sistema/PA(t)/Derivative  
Sistema/PA(t)/Add (algebraic variable)

```
plotsignals(x1.t,x1.Pao,x1.P0,x1.P1,x1.P2,x1.PID,Signal)
```

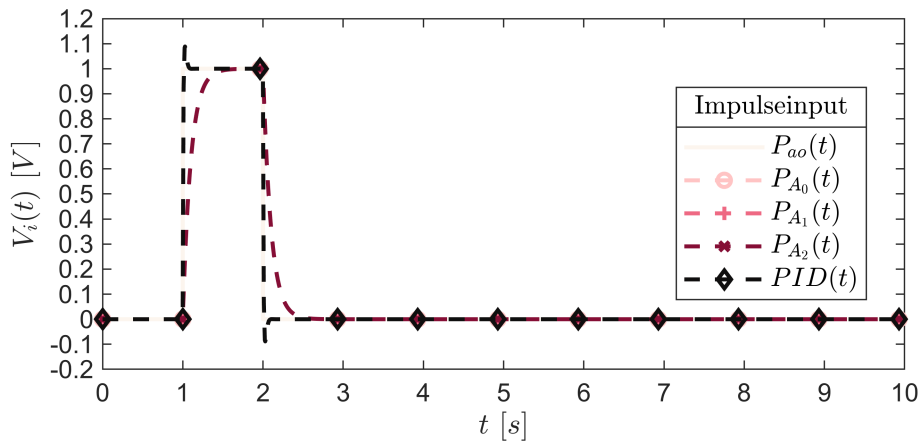


## Respuesta al impulso

```
Signal = 'Impulse';  
set_param('Sistema/S1','sw','0');  
set_param('Sistema/Pao(t)','sw','1');  
x2 = sim(file,parameters);
```

Found algebraic loop containing:  
Sistema/PA(t)/Gain  
Sistema/PA(t)/Gain1  
Sistema/PA(t)/Derivative  
Sistema/PA(t)/Add (algebraic variable)

```
plotsignals(x2.t,x2.Pao,x2.P0,x2.P1,x2.P2,x2.PID,Signal)
```

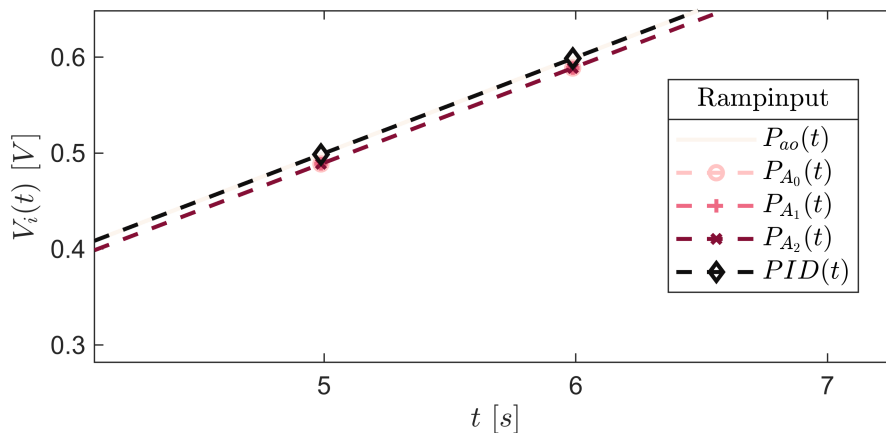


## Respuesta a la rampa

```
Signal = 'Ramp';
set_param('Sistema/S2','sw','1');
set_param('Sistema/Pao(t)','sw','0');
x3 = sim(file,parameters);
```

Found algebraic loop containing:  
 Sistema/PA(t)/Gain  
 Sistema/PA(t)/Gain1  
 Sistema/PA(t)/Derivative  
 Sistema/PA(t)/Add (algebraic variable)

```
plotsignals(x3.t,x3.Pao,x3.P0,x3.P1,x3.P2,x3.PID,Signal)
```

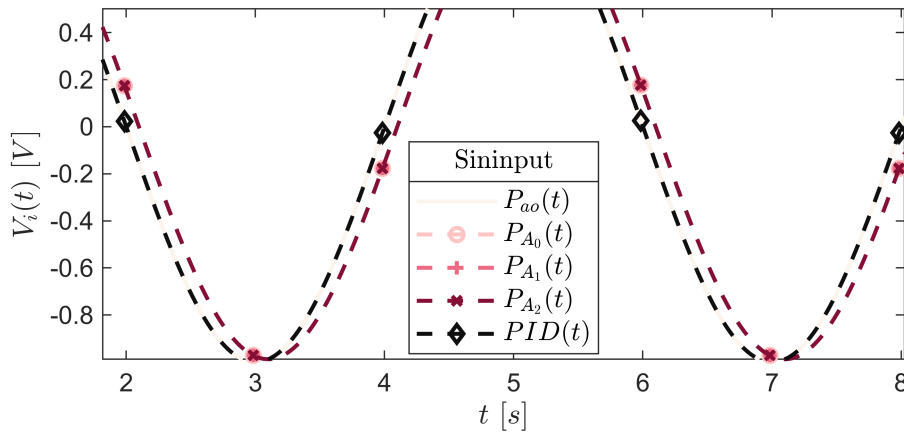


## Respuesta a la función sinusoidal

```
Signal = 'Sin';
set_param('Sistema/S2','sw','0');
set_param('Sistema/Pao(t)','sw','0');
x4 = sim(file,parameters);
```

Found algebraic loop containing:  
 Sistema/PA(t)/Gain  
 Sistema/PA(t)/Gain1

```
plotsignals(x4.t,x4.Pao,x4.P0,x4.P1,x4.P2,x4.PID,Signal)
```



## Funcion: Respuesta a las seales

```
function plotsignals(t,Pao,P0,P1,P2,PID,Signal)
    set(fgure(),'color','w')
    set(gcf,'units','centimeters','position',[1,1,18,8])
    set(gca,'FontName','Times New Roman','FontSize',11)
    colors = [252, 245, 238;
              255, 196, 196;
              238, 105, 131;
              133, 14, 53;
              15, 14, 14]/255;
    colororder(colors)
    plot(t,Pao,'-',t,P0,'--o',t,P1,'--+',t,P2,'--x',t,PID,'--d',...
        'LineWidth',1.5,'MarkerSize',5,'MarkerIndices',1:1000:length(t));
    L = legend('$P_{ao}(t)$','$P_{A_0}(t)$','$P_{A_1}(t)$','$P_{A_2}(t)$','$PID(t)$');
    set(L,'Interpreter','Latex','FontSize',10,'location','best','box','on')
    title(L,[Signal,'input'],'FontSize',10)
    xlabel('$t$ [s]','Interpreter','Latex','FontSize',11)
    ylabel('$V_i(t)$ [V]','Interpreter','Latex','FontSize',11)
    if Signal == "Step"
        xlim([0,10]); xticks(0:1:10)
        ylim([0,1.2]); yticks(0:0.1:1.2)
    elseif Signal == "Impulse"
        xlim([0,10]); xticks(0:1:10)
        ylim([-0.2,1.2]); yticks(-0.2:0.1:1.2)
    elseif Signal == "Ramp"
        xlim([0,10]); xticks(0:1:10)
        ylim([-0.05,1.1]); yticks(0:0.1:1.2)
    elseif Signal == "Sin"
        xlim([0,10]); xticks(0:1:10)
```

```
        ylim([-1.2,1.2]); yticks(-1.2:0.2:1.2)
end
exportgraphics(gcf,[Signal,'.pdf'],'ContentType','vector')
% exportgraphics(gcf,[Signal,'.png'],'Resolution',600);
% print(Signal,'-dsvg','-r600');
% print(Signal,'-depsc','-r600')

end
```